

Evidence-Grounded Task World Model Induction for LLM Agents

Working outline for an IntentCap follow-up

Agents act on text, but tasks unfold in worlds.

We study how agents reconstruct an explicit, executable, evidence-grounded model of the task world from heterogeneous, incomplete, and conflicting inputs before acting.

1. Problem

IntentCap constrains how an agent may act once task-relevant intent, authority fields, and action scope are known. The follow-up asks a prior question: **what task world has the agent reconstructed in the first place?**

In current agents, that model is mostly implicit inside a mixed context window. Instructions, documents, tables, tool outputs, code, logs, and prior observations are consumed together, then the model acts directly from text. The missing layer is an explicit representation of the entities, states, relations, transitions, and constraints that actually govern the task.

heterogeneous evidence $\rightarrow$ task-world model $\rightarrow$ decision / action

2. Core abstraction: Task World Model (TWM)

A TWM is a task-scoped, explicit model of the world needed to decide and act; it is not a general physical simulator. A minimal representation is:

$$W_{\tau} = (E, S, R, T, C, U)$$

E - entities: objects, actors, resources	S - states: current attributes / status
R - relations: dependencies and links	T - transitions: how actions/events change state
C - constraints: rules, eligibility, invariants	U - unknowns: unresolved or missing facts

Every element also carries provenance and epistemic status: **observed** / **derived** / **assumed** / **unknown**. This is essential: the model should preserve what is not known rather than silently completing it.

3. Evidence-grounded induction

The input is not a single document but a heterogeneous evidence set D : user instructions, policies, documents, spreadsheets, tool results, logs, code, and historical cases. The induction problem is:

$$D \rightarrow W_{\tau}, \text{ with } \text{provenance}(x) = (\text{source}, \text{evidence}, \text{status})$$

- Resolve the same entity across sources without collapsing distinct objects.
- Recover task-relevant state, relations, rules, and transitions rather than merely summarizing text.
- Represent conflict explicitly when sources disagree.
- Keep missing information as UNKNOWN unless a justified derivation or explicit assumption closes it.
- Produce a representation that downstream decisions and verifiers can actually consume.

4. Main failure modes

Missing-state failure: A task-relevant state variable never enters the model.

Wrong-relation failure: Entities are present, but their dependency or ownership relation is wrong.

Unsupported inference: An assumption or weak inference is promoted to fact.

Conflict collapse: Conflicting sources are silently merged or one is chosen without justification.

Unknown collapse: The model fills an unresolved gap because a complete answer looks more coherent.

Transition failure: The model describes the current world but gets how actions change it wrong.

5. Research questions

- RQ1 - Can LLM agents reconstruct accurate task world models from heterogeneous, incomplete, and conflicting evidence?
- RQ2 - Which components fail most often: entities/states, relations, constraints, transitions, provenance, or unknown handling?
- RQ3 - Does making the task world explicit improve downstream decisions compared with direct context-to-action prompting?
- RQ4 - How much does epistemic labeling (observed/derived/assumed/unknown) reduce unsupported decisions and silent conflict resolution?

6. Evaluation design

Each benchmark instance contains evidence D , a gold task-world model W^* , and downstream decision/action queries. Evaluation has two levels:

- Model reconstruction: component-level precision/recall or graph/state agreement, provenance correctness, conflict detection, and UNKNOWN preservation.
- Decision utility: compare direct text -> action against text -> explicit TWM -> action on the same evidence; measure downstream decision fidelity and task success.

Controlled perturbations should deliberately vary missing evidence, contradictory sources, stale observations, and tempting but unsupported completions. The key result should not be only “extraction accuracy,” but whether the induced model changes what the agent does.

7. Relationship to IntentCap

IntentCap: Who is allowed to determine or influence an action?

This work: What task world is the agent reasoning and acting over, and what evidence supports it?

Evidence -> Task World Model -> Intent / Authority -> Action

IntentCap constrains action inside a task world. This paper makes the task world itself explicit, evidence-grounded, and auditable. A later behavioral-closure paper can ask whether the world after execution actually satisfies the user's goal.

8. Expected contribution

- A precise abstraction for an explicit, task-scoped world model in LLM-agent systems.
- A provenance-aware induction method that preserves observed, derived, assumed, conflicting, and unknown information.
- A benchmark/harness that tests reconstruction under heterogeneous evidence and measures downstream behavioral impact.
- Evidence that explicit task-world reconstruction can separate “reasoning coherently over text” from “reasoning correctly about the task world.”

Scope discipline: this paper is about reconstructing the task world before action, not yet about general semantic-vs-behavioral closure.